

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT : SCALA FOR DATA SCIENCE

Practical No : 11 and 12

Aim 11 : Perform a group-by operation in Spark DataFrames to compute the average of each group.

Aim 12 : Join two CSV files in Spark DataFrames based on a common column and write the output to a file.

CODE 11 :

```
from pyspark.sql import SparkSession
from pyspark.sql.functions import avg
```

```
spark = SparkSession.builder \
    .appName("GroupByAverage") \
    .master("local[*]") \
    .getOrCreate()
```

```
df = spark.read.csv(
    "students.csv",
    header=True,
    inferSchema=True
)
```

```
print("Original Data:")
df.show()
```

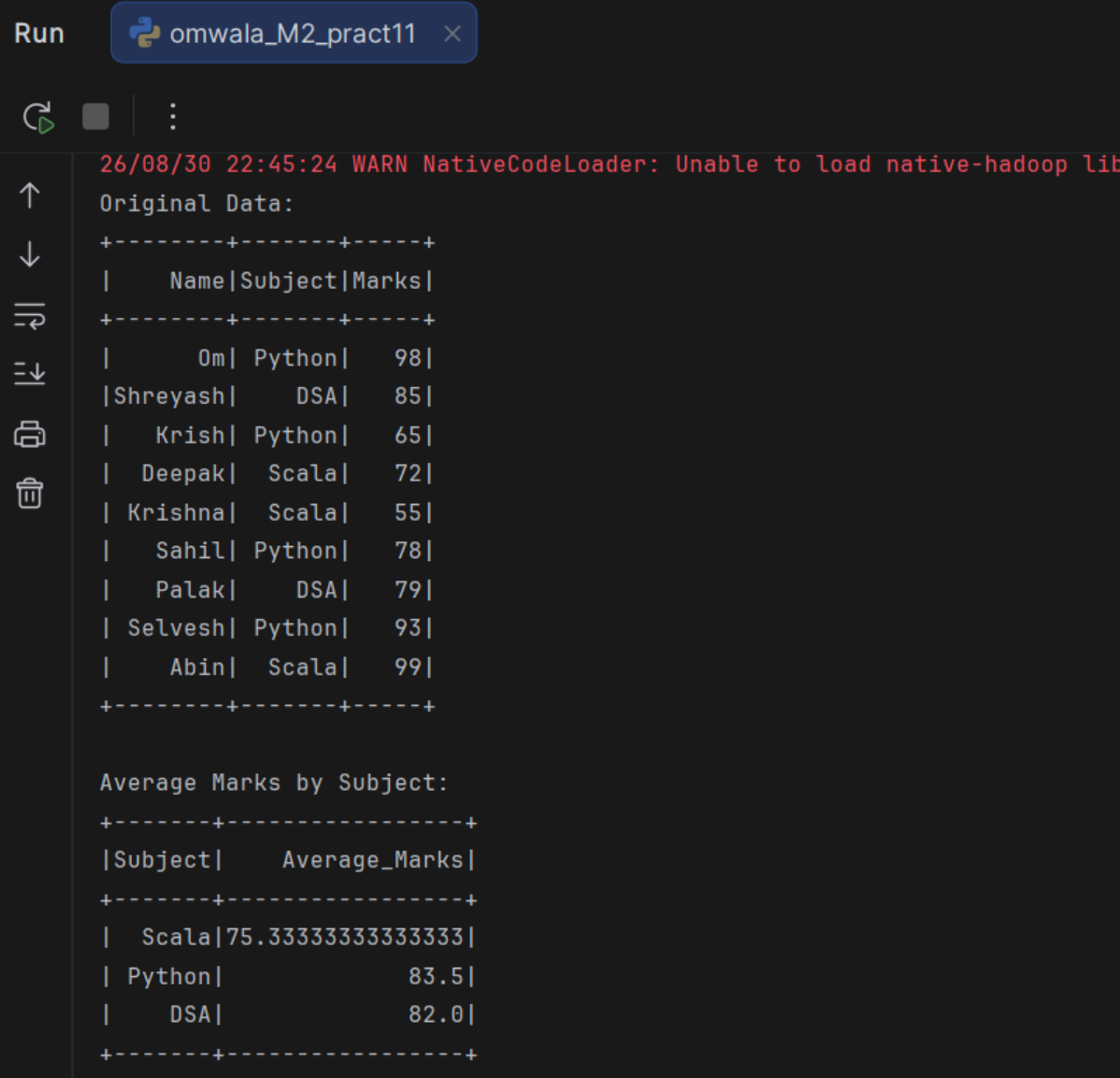
```
result = df.groupBy("Subject").agg(
    avg("Marks").alias("Average_Marks")
)
```

```
print("Average Marks by Subject:")
result.show()
```

```
spark.stop()
```

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT : SCALA FOR DATA SCIENCE

OUTPUT :



```
Run omwala_M2_pract11 x
26/08/30 22:45:24 WARN NativeCodeLoader: Unable to load native-hadoop lib
Original Data:
+-----+-----+-----+
|   Name|Subject|Marks|
+-----+-----+-----+
|    Om| Python|  98|
|Shreyash|   DSA|  85|
|   Krish| Python|  65|
|  Deepak|  Scala|  72|
| Krishna|  Scala|  55|
|   Sahil| Python|  78|
|   Palak|   DSA|  79|
| Selvesh| Python|  93|
|    Abin|  Scala|  99|
+-----+-----+-----+

Average Marks by Subject:
+-----+-----+
|Subject| Average_Marks|
+-----+-----+
|  Scala|75.33333333333333|
| Python|          83.5|
|   DSA|          82.0|
+-----+-----+
```

CODE 12 :

```
from pyspark.sql import SparkSession

print("Om Wala S119")
spark = SparkSession.builder \
    .appName("FIFA 2026 CSV Join") \
    .master("local[*]") \
    .getOrCreate()

teams_df = spark.read.csv(
    "fifa_2026_teams.csv",
    header=True,
    inferSchema=True
)
```

Name : Om Wala
Roll No : S119

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT : SCALA FOR DATA SCIENCE

```
stats_df = spark.read.csv(  
    "fifa_2026_stats.csv",  
    header=True,  
    inferSchema=True  
)
```

```
print("\nFIFA 2026 Teams:")  
teams_df.show()
```

```
print("\nFIFA 2026 Stats:")  
stats_df.show()
```

```
joined_df = teams_df.join(  
    stats_df,  
    on="team_code",  
    how="inner"  
)
```

```
print("\nJoined FIFA 2026 Data:")  
joined_df.show()
```

```
joined_df.write \  
    .mode("overwrite") \  
    .option("header", True) \  
    .csv("fifa_2026_joined_output")
```

```
print("\nJoined data successfully saved!")
```

```
spark.stop()
```

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT : SCALA FOR DATA SCIENCE

OUTPUT :

```
Run omwala_M2_pract12 x
```

```
26/08/30 22:36:51 WARN NativeCodeLoader: Unable to load native-hadoop library for your platf
```

```
FIFA 2026 Teams:
```

team_code	team_name	confederation	star_player
ARG	Argentina	CONMEBOL	Lionel Messi
BRA	Brazil	CONMEBOL	Vinicius Junior
FRA	France	UEFA	Kylian Mbappe
ESP	Spain	UEFA	Lamine Yamal
ENG	England	UEFA	Jude Bellingham
GER	Germany	UEFA	Jamal Musiala
POR	Portugal	UEFA	Cristiano Ronaldo
JPN	Japan	AFC	Takefusa Kubo
MAR	Morocco	CAF	Achraf Hakimi
USA	United States	CONCACAF	Christian Pulisic

```
Run omwala_M2_pract12 x
```

```
FIFA 2026 Stats:
```

team_code	matches_won	attack_rating	total_goals
ARG	7	78	19
BRA	3	82	10
FRA	6	80	20
ESP	7	79	12
ENG	6	76	20
GER	2	75	11
POR	2	77	8
JPN	1	72	8
MAR	4	70	10
USA	3	68	11

Name : Om Wala
Roll No : S119

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT : SCALA FOR DATA SCIENCE

Run omwala_M2_pract12 x

Joined FIFA 2026 Data:

team_code	team_name	confederation	star_player	matches_won	attack_rating	total_goals
ARG	Argentina	CONMEBOL	Lionel Messi	7	78	19
BRA	Brazil	CONMEBOL	Vinicius Junior	3	82	10
FRA	France	UEFA	Kylian Mbappe	6	80	20
ESP	Spain	UEFA	Lamine Yamal	7	79	12
ENG	England	UEFA	Jude Bellingham	6	76	20
GER	Germany	UEFA	Jamal Musiala	2	75	11
POR	Portugal	UEFA	Cristiano Ronaldo	2	77	8
JPN	Japan	AFC	Takefusa Kubo	1	72	8
MAR	Morocco	CAF	Achraf Hakimi	4	70	10
USA	United States	CONCACAF	Christian Pulisic	3	68	11